ITA0329-Mobile Computing

Co1-Assessment 2

Short Answer Questions

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1.Given:

- spectrum = 45 MHz = 45000 kHz
- Bandwidth = 25 kHz

Formula:

N=Total spectrum/2*Bandwidth

N=45000/2*25

N=45000/50

N=900

900 full-duplex voice channels.

2. Given:

- Total voice channels = 560
- Cluster size = 7 cells

N=Total voice channels/cluster size
N=560/7
N=80

80 channels a single cell will be allocated.

3. Given:

- carriers = 100
- Time slots = 8
- Carrier spacing = 200 kHz

N=carriers*Time slots
N=100*8
N=800
800 simultaneous voice channels.

4.Given:

- Base station = 80 W
- Handset Transmit = 4 W

P{dB}=10log{10}*(p1/p2)

P{dB}=10 log {10}*20

P{dB}=13.01dB

5. Given:

- Carrier power = −60 dBm
- Channel Interference = −92 dBm

Formula:

C/I = PC-PI
= -60 - (-92)
= 32 dB

6. Given:

- spectrum = 42 MHz = 42,000 kHz
- Bandwidth = 300 kHz
- Cluster = 7

Formula:

N=(Total bandwidth/Channel bandwidth) / Cluster size
N=42000/300=140

N=140/7=20

9.Given:

- Cluster size = 4 cells
- Channels per cell = 15

Formula:

N=Cluster size*Channels per cell
N=4*15=60

11.Given:

- TDMA frame duration = 5 ms
- Number of time slots = 8

Formula:

T=Frame duration / Number of slots
T= 5 / 8= 0.625 ms

12.Given:

- Total bandwidth = 24 MHz = 24,000 kHz
- Carrier spacing = 200 kHz

Formula:

N = Total bandwidth / Carrier spacing
N= 24000 / 200= 120

15. Given:

- Number of cells = 6
- Calls supported per cell = 250

Formula:

N=Number of cells * Calls per cell
N= 6 * 250 = 1500

17. Given:

- Number of channels = 150
- Bandwidth per channel = 25 kHz

Formula:

B=N * B{channel}
B=150 * 25=3750 kHz
B=3.75MHz

18. Given:

- Number of adjacent frequency bands = 30
- Guard band between each = 40 kHz

Formula:

B=N * B{each guard}

Assuming 30 guard bands:

B = 30 * 40
= 120 kHz = 1.2 MHz

20. Given:

- Regional centers = 8
- Base stations per center = 40
- Users per base station = 250

Formula:

N=R*B*U
N=8*40*250 \]
N=80,000

21.Given:

- Total area = 400 km^2
- Cell radius = 400 m = 0.4 km
- Approximate cell area = R^2

Formula:

N = Total area / Area of one cell
A = (0.4)^2 = 0.16 km^2
N = 400 /0.16 = 2500

22. Given:

- Total spectrum = 6 MHz = 6,000 kHz
- Bandwidth per channel = 30 kHz

Formula:

N= Total bandwidth / Bandwidth per channel
N=6000 / 30 = 200

23.Given:

- Number of subscribers = 1,200
- Calls per subscriber per hour = 4
- Average call duration = 2.5 minutes

Formula:

A=N* C*T/60
A=1200* 4 * 2.5/60
A=12000/ 60=200

24 Given:

- Total bandwidth = 40 MHz = 40,000 kHz
- Bandwidth required per user = 400 kHz

Formula:

N=Total bandwidth/Bandwidth per user
N=40000 / 400 = 100

25.Given:

- Number of base stations = 1,200
- Capacity per base station = 600 users

Formula:

N=Number of stations*Users per station
N=1200*600=720000

26. Given:

- Total cells = 200
- Cells searched per round = 4

Formula:

N=Total cells/Cells per round

N=200 / 4 = 50

28.Given:

- Total subscribers = 600,000
- Percentage roaming = 15%

Formula:

N=Percentage/100 * Total subscribers

Calculation:

N=15 / 100* 600000
N=90000

29. Given:

- Maximum capacity = 12,000 calls/second
- Current load = 9,000 calls/second

Formula:

U=Current load/Maximum capacity*100

U=9000/12000*100

U=75%

30.Given:

- Total spectrum = 48 MHz = 48,000 kHz
- Bandwidth per channel = 200 kHz
- Frequency reuse factor = 4

Formula:

N = (Total bandwidth/Channel bandwidth)/Reuse factor
N = 48000/200 = 240
N = 240 / 4 = 60

32. Given:

- Total handoff attempts = 2,500
- Success rate = 99%

Formula:

N=Success rate/100*Total attempts
N=99/100*2500
N=2475

34. Given:

- Number of cells = 7
- Users supported per cell = 900

Formula:

N=Number of cells*Users per cell

N=7*900=6300